Last Time:

- $V = \text{span}(S)$ w/ $\#S < \infty \Rightarrow \exists$ (fin) basis $\beta \subseteq S$ of V.
- Replacement Thm: $V = \text{span}(G)$, $\#G = n < \infty$, $B \subseteq V$ l.i. w/ $\#B = m < \infty$. Then $m \leq n$ and $\exists H \subseteq G$ w/ $\#H = n - m$ s.t. $B \cup H$ spans V.
- Cor 1.55: β fin basis of a v.s. V. Then $\forall$ basis $\tilde{\beta}$ of V, $\#\beta = \#\tilde{\beta}$.

Today:

Continue §1.4 Basis and Dimension (Dimension of vector spaces)

Cor 1.55 leads to def of dimension:

Def 1.56: V v.s. is finite-dimensional (f.d.) if $\exists$ fin basis. In this case, if β is a fin basis, then dimension (dim) of V is def by $\dim V := \#\beta$.

well-def b/c of Cor 1.55 ↑

(only deps on V, not on the choice of β)

Def 1.57: V v.s. is infinite-dimensional (∞-dim'l) if $\nexists$ a fin basis for V. In this case, dim of V is $\dim V := \infty$ (V has a ∞-basis β, so $\#\beta = \infty$)

Ex 1.58:

(1) $\dim\{0\} = 0$ b/c $\emptyset$ is a basis.

(2) $\dim \mathbb{R}^n = n$ b/c of the std basis $(e_1, \dots, e_n)$.

(3) $\dim \mathbb{R}[x]_{\le n} = n+1$ b/c of the std basis $(1, x, x^2, \dots, x^n)$.

(4) $\dim \mathbb{R}[x] = \infty$

Pf: Let $\beta := \{x^n : n \in \mathbb{Z}_{\ge 0}\}$ a basis for $\mathbb{R}[x]$ by Ex 1.47. If $\exists$ fin basis γ for $\mathbb{R}[x]$, then Cor 1.55 says that $\infty > \#\gamma = \#\beta = \infty$, ※. Thus $\mathbb{R}[x]$ is not f.d. $\Rightarrow \dim \mathbb{R}[x] = \infty$. □

(5) $\dim M_{m\times n}(\mathbb{R}) = mn$.

Pf: Exercise (show that $\beta = \{E_{ij} : 1 \le i \le m,\ 1 \le j \le n\}$ is a basis, where $E_{ij} \in M_{m\times n}(\mathbb{R})$ def by $(E_{ij})_{kl} = \delta_{ik}\cdot\delta_{jl}$.) □

(In $M_{2\times 2}(\mathbb{R})$, $E_{11} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $E_{21} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, …)

Cor 1.59: Say $\dim V = n < \infty$.

(1) If $V = \text{span}(S)$ w/ $\#S = n$, then S is a basis.

(2) $\gamma \subseteq V$ l.i. w/ $\#\gamma = n \Rightarrow \gamma$ is a basis.

(3) (Basis Extension Thm (BET)): $\gamma \subseteq V$ l.i., then $\exists \beta \supseteq \gamma$ in V s.t. β is a basis for V.

Pf: (1): $V = \text{span}(S) \Rightarrow \exists$ basis $\beta \subseteq S$ for V by Prop 1.49.

If S l.d., then $\beta \subsetneq S$ by def. S fin

$\Rightarrow n = \#\beta < \#S = n$, ※. (Cor 1.55, S fin) Thus, S l.i. & spans V

$\Rightarrow S$ is a basis. ▨

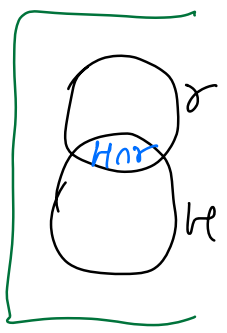
(2): Suppose that γ is not a basis (i.e., $\text{span}(\gamma) \subsetneq V$). Then, $\exists v \in V \setminus \text{span}(\gamma)$

$\Rightarrow \gamma \cup \{v\}$ l.i. by Lem 1.44. $\dim V = n$ says $\exists$ basis β (also spans V) w/ $\#\beta = n$. But $\gamma \cup \{v\}$ l.i. w/ $\#(\gamma \cup \{v\}) = n+1$, ※ to Replacement Thm ($\#(\gamma \cup \{v\}) \leq \#\beta$). Thus, γ is a basis. □

(3): Say $G \subseteq V$ is a basis $\Rightarrow \#G = \dim V = n$ by def.

By Replacement Thm, $\#\gamma \leq \#G = n$ and $\exists H \subseteq G$ w/ $\#H = n - \#\gamma$ s.t. $\gamma \cup H$ spans V. (*) So

$$\#(\gamma \cup H) = \#\gamma + \#H - \#(\gamma \cap H) \leq \#\gamma + \#H$$
$$= \#\gamma + (n - \#\gamma) = n$$



If $\#(\gamma \cup H) = n$, then $\beta := \gamma \cup H$ is a basis by (1).

If $\#(\gamma \cup H) < n$, then by Prop 1.49 & (*), $\exists$ basis

$\alpha \subseteq \gamma \cup H \Rightarrow n = \#\alpha \leq \#(\gamma \cup H) < n$, ⁂.

Cor 1.55

Thus, $\beta := \gamma \cup H \supseteq \gamma$ is a basis for V. □

Ex 1.51 Revisited:

$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\} \subseteq \mathbb{R}^3$ $\dim = 3$ by Ex 1.58(2)

Thanks to BET & Cor 1.55, S is l.d.

When we saw that $T = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \right\}$ l.i.,

then $\#T = 3 = \dim \mathbb{R}^3 \Rightarrow T$ basis for $\mathbb{R}^3$

by Cor 1.59(2).

This significantly simplifies Ex 1.51

↑ results about dim / basis

Q: What can we say about basis / dim of subsp W of a v.s. V when we know basis / dim of V?

Thm 1.60: $W \subseteq V$ subsp, V f.d. Then,

(1) $\dim W \leq \dim V$ (so W f.d.)

(2) If $\dim W = \dim V$, then $W = V$.

Pf: (1): By Prop 1.49 & a Rmk, $\exists$ basis β of W.

Then $\beta \subseteq V$ l.i. by def, so BET says $\exists \gamma \supseteq \beta$ and $\gamma \subseteq V$ basis. So

$$\dim W = \#\beta \leq \#\gamma = \dim V < \infty.$$ ▨

(2): If $\dim W = \dim V$, then take a basis β of W.

Then β l.i. and $\#\beta = \dim W = \dim V$

$\Rightarrow \beta$ basis of V by Cor 1.59(2).

$\Rightarrow W = \text{span}(\beta) = V$. □

Ex 1.61: Recall that

$$W := \left\{ \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R}^n : \sum_{i=1}^{n} a_i = 0 \right\} \subsetneq \mathbb{R}^n \quad \text{subsp.}$$

Then $\dim W < \dim \mathbb{R}^n = n$ by Thm 1.60.

Goal: Compute $\dim W$.

Consider $\beta := \underbrace{\{e_i - e_{i+1} : 1 \leq i < n\}}_{\{e_1 - e_2, e_2 - e_3, e_3 - e_4, \dots, e_{n-1} - e_n\}} \subseteq W$, where $(e_1, \dots, e_n)$ std basis.

Claim: β l.i.

Pf: Say $\sum_{i=1}^{n-1} a_i(e_i - e_{i+1}) = 0$. Then

$$0 = \sum_{i=1}^{n-1} a_i(e_i - e_{i+1})$$

$$= a_1 e_1 + (a_2 - a_1)e_2 + (a_3 - a_2)e_3 + \cdots + (a_{n-1} - a_{n-2})e_{n-1} - a_{n-1} e_n$$

$\Rightarrow a_1 = 0,\ a_{n-1} = 0,\ a_{i+1} - a_i = 0 \quad \forall\ 1 \le i \le n-2$

b/c $e_1, \ldots, e_n$ l.i.

$\Rightarrow a_i = 0 \quad \forall\ 1 \le i \le n-1$. Thus, β l.i. $\square$

Note that $\#\beta = n-1$, so BET

$\Rightarrow \exists\ \gamma \supseteq \beta$ basis of W w/

$n-1 = \#\beta \le \#\gamma = \dim W < n$

$\Rightarrow \gamma = \beta$ basis of W and $\dim W = \boxed{n-1}$.